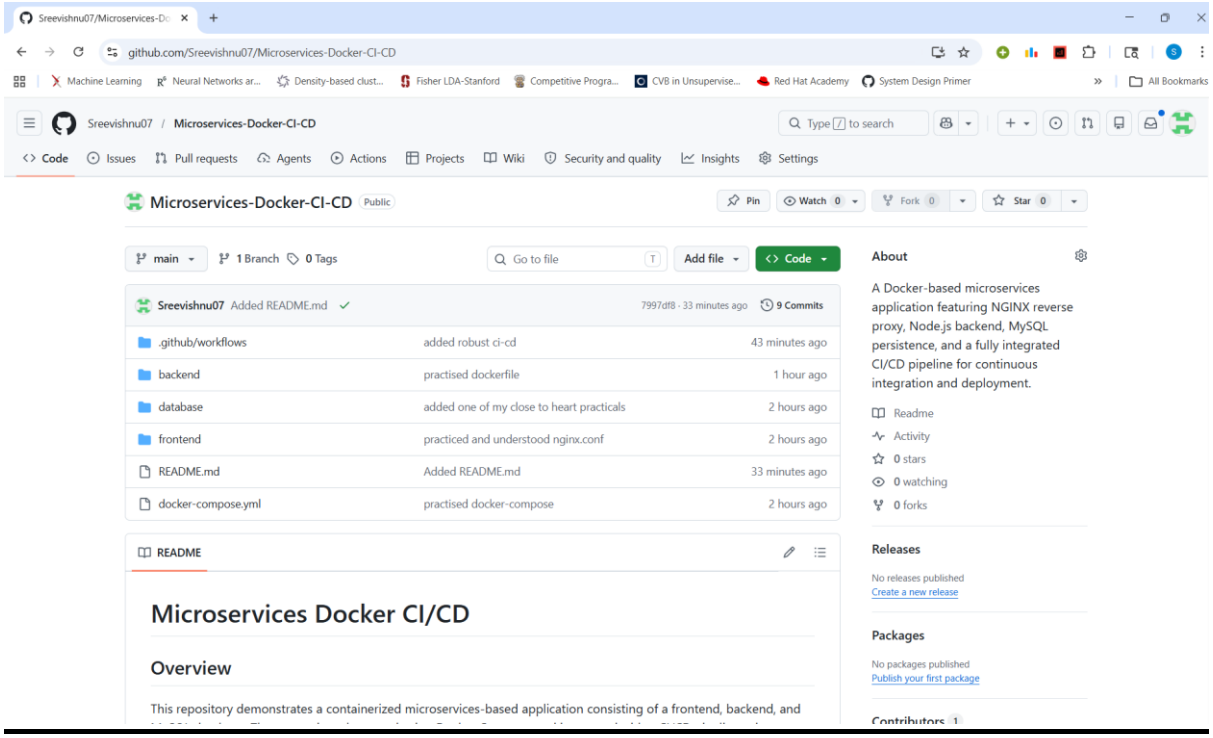
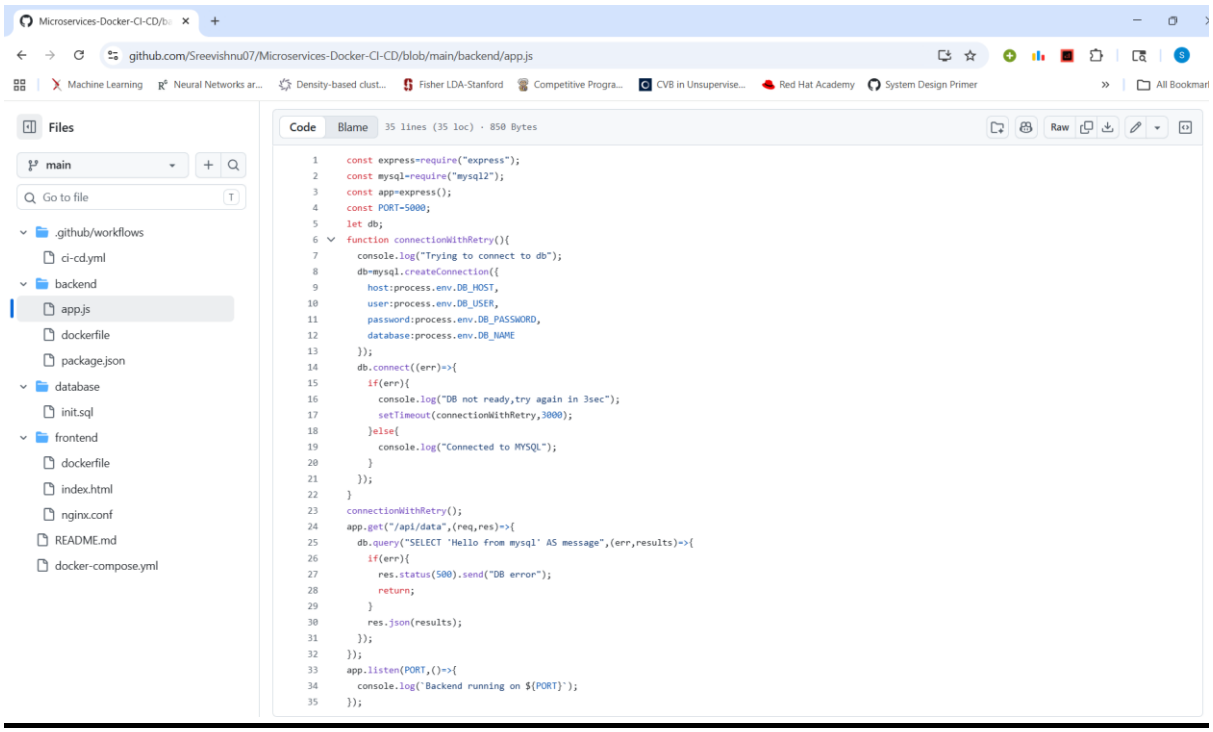
Design and Implementation of Docker Microservices with CI/CD Integration





github.com/Sreevishnu07/Microservices-Docker-CI-CD/blob/main/backend/dockerfile

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Files

main

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- README.md
- docker-compose.yml

Microservices-Docker-CI-CD / backend / dockerfile

Sreevishnu07 practised dockerfile

Code | Blame | 7 lines (7 loc) · 103 Bytes

```
FROM node:18
WORKDIR /app
COPY package.json .
RUN npm install
COPY . .
EXPOSE 5000
CMD ["npm","start"]
```

---

github.com/Sreevishnu07/Microservices-Docker-CI-CD/blob/main/backend/package.json

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Microservices-Docker-CI-CD / backend / package.json

Sreevishnu07 added one of my close to heart practicals | 216ae3f · 3 hours ago | History

Code | Blame | 13 lines (13 loc) · 229 Bytes | Raw

```
{
"name":"backend-service",
"version":"1.0.0",
"description":"simple backend api for microservices demo",
"main":"app.js",
"scripts":{
  "start":"node app.js"
  },
"dependencies":{
  "express":"^4.18.2",
  "mysql2":"^3.6.0"
  }
}
```

Microservices-Docker-CI-CD/fr

github.com/Sreevishnu07/Microservices-Docker-CI-CD/blob/main/frontend/dockerfile

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Microservices-Docker-CI-CD / frontend / dockerfile

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Code | Blame | 5 lines (5 loc) · 142 Bytes

```
FROM nginx:latest
RUN rm /etc/nginx/conf.d/default.conf
COPY nginx.conf /etc/nginx/nginx.conf
COPY index.html /usr/share/nginx/html
EXPOSE 80
```

Microservices-Docker-CI-CD/fr

github.com/Sreevishnu07/Microservices-Docker-CI-CD/blob/main/frontend/index.html

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Microservices-Docker-CI-CD / frontend / index.html

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Code | Blame | 12 lines (12 loc) · 192 Bytes | Raw

```
<!DOCTYPE html>
<html>
<head>
  <title>Simple microservices app</title>
</head>
<body>
  <h2>Click to get data</h2>
  <a href="/api/data">
     <button>Get data</button>
  </a>
</body>
</html>
```

Microservices-Docker-CI-CD/fr

github.com/Sreevishnu07/Microservices-Docker-CI-CD/blob/main/frontend/nginx.conf

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Microservices-Docker-CI-CD / frontend / nginx.conf

Sreevishnu07 practiced and understood nginx.conf

Code | Blame | 13 lines (13 loc) · 190 Bytes

```
events{}
http{
  server{
    listen 80;
    location /{
      root /usr/share/nginx/html;
      index index.html;
    }
    location /api/{
      proxy_pass http://backend:5000;
    }
  }
}
```

Microservices-Docker-CI-CD/d

github.com/Sreevishnu07/Microservices-Docker-CI-CD/blob/main/docker-compose.yml

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Code | Blame | 37 lines (35 loc) · 653 Bytes | Raw

```
services:
  database:
    image: mysql:8
    container_name: mysqldb
    restart: always
    environment:
      MYSQL_ROOT_PASSWORD: root
      MYSQL_DATABASE: mydb
    ports:
      - "3307:3306"
    volumes:
      - db_data:/var/lib/mysql

  backend:
    build: ./backend
    container_name: backend
    restart: always
    depends_on:
      - database
    environment:
      DB_HOST: database
      DB_USER: root
      DB_PASSWORD: root
      DB_NAME: mydb
    ports:
      - "5000:5000"

  frontend:
    build: ./frontend
    container_name: frontend
    restart: always
    depends_on:
      - backend
    ports:
      - "80:80"
```

Microservices-Docker-CI-CD/.g

github.com/Sreevishnu07/Microservices-Docker-CI-CD/blob/main/.github/workflows/ci-cd.yml

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Microservices-Docker-CI-CD / .github / workflows / ci-cd.yml

View Runs

Sreevishnu07 added robust ci-cd ✓ 8e120ba · 46 minutes ago History

Code Blame 88 lines (71 loc) · 2.09 KB Raw

```yaml
name: Full CI-CD pipeline

on:
  push:
    branches:
      - main

jobs:
  test:
    runs-on: ubuntu-latest

    services:
      mysql:
        image: mysql:8
        env:
          MYSQL_ROOT_PASSWORD: root
          MYSQL_DATABASE: my_db
        ports:
          - "3306:3306"
        options: >-
          --health-cmd="mysqladmin ping -h localhost -proot"
          --health-interval=5s
          --health-timeout=5s
          --health-retries=5

    steps:
      - name: Checkout code
        uses: actions/checkout@v4

      - name: Setup Node
        uses: actions/setup-node@v4
```

Microservices-Docker-CI-CD/.g

github.com/Sreevishnu07/Microservices-Docker-CI-CD/blob/main/.github/workflows/ci-cd.yml

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Microservices-Docker-CI-CD / .github / workflows / ci-cd.yml

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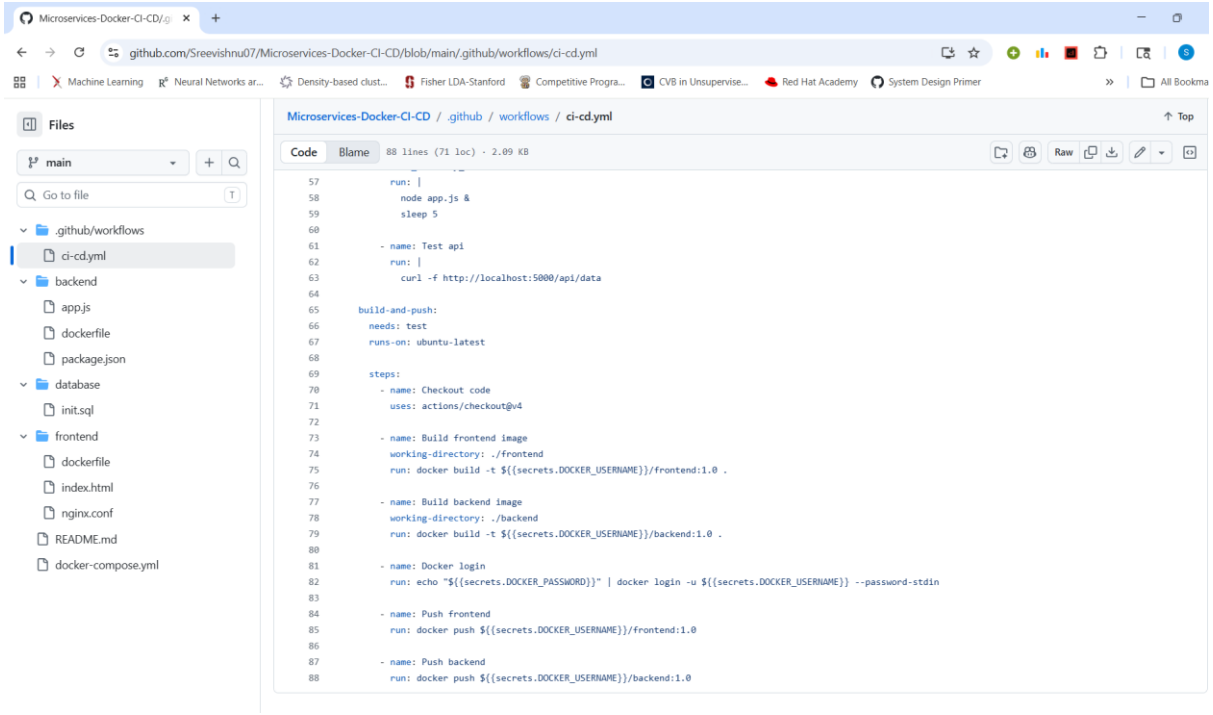
Code Blame 88 lines (71 loc) · 2.09 KB Raw

```yaml
      - name: Setup Node
        uses: actions/setup-node@v4
        with:
          node-version: 18

      - name: Install backend dependencies
        working-directory: ./backend
        run: npm install

      - name: Install MySql Client
        run: sudo apt-get update && sudo apt-get install -y mysql-client

      - name: Wait for MySql
        run: |
          for i in {1..10}; do
            mysqladmin ping -h 127.0.0.1 -proot && break
            echo "Waiting for mysql.."
            sleep 3
          done

      - name: Run Backend
        working-directory: ./backend
        env:
          DB_HOST: 127.0.0.1
          DB_USER: root
          DB_PASSWORD: root
          DB_NAME: my_db
        run: |
          node app.js &
          sleep 5

      - name: Test api
        run: |
          curl -f http://localhost:5000/api/data
```


Microservices-Docker-CI-CD/
github.com/Sreevishnu07/Microservices-Docker-CI-CD/blob/main/.github/workflows/ci-cd.yml
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Microservices-Docker-CI-CD / .github / workflows / ci-cd.yml
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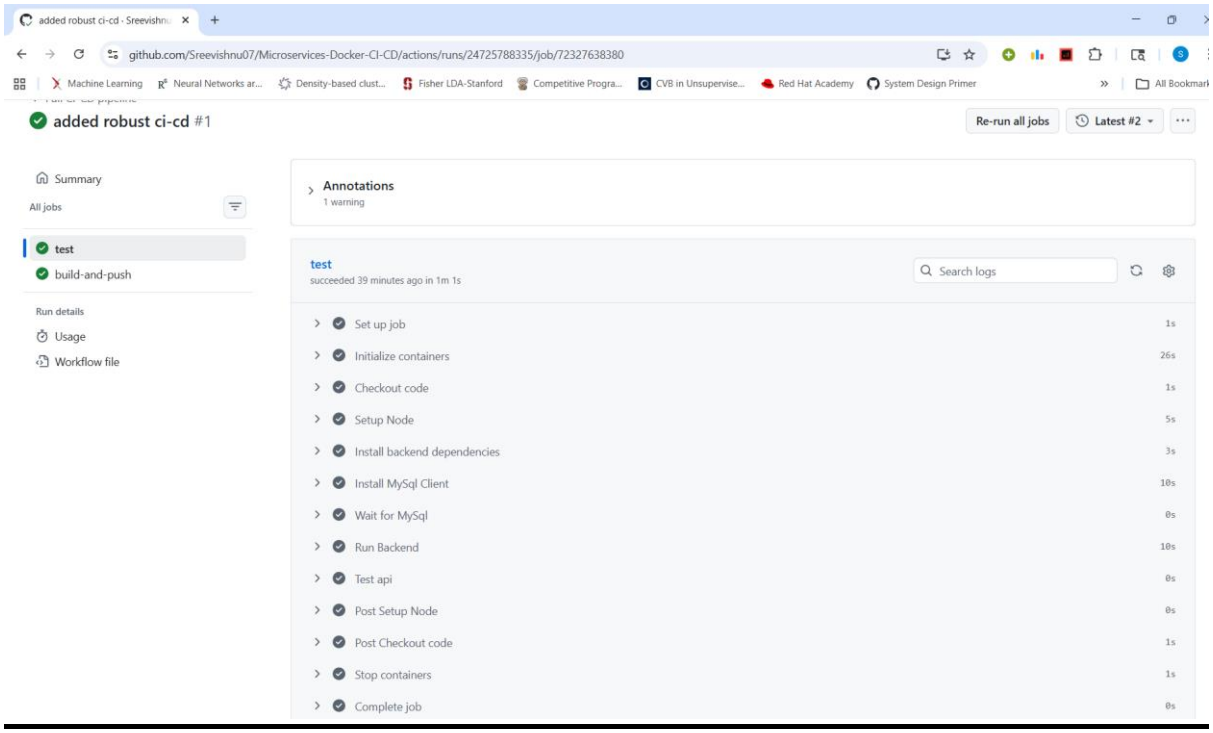
Code
Blame
88 lines (71 loc) · 2.09 KB
Raw
57 run: |
58 node app.js &
59 sleep 5
60
61 - name: Test api
62 run: |
63 curl -f http://localhost:5000/api/data
64
65 build-and-push:
66 needs: test
67 runs-on: ubuntu-latest
68
69 steps:
70 - name: Checkout code
71 uses: actions/checkout@v4
72
73 - name: Build frontend image
74 working-directory: ./frontend
75 run: docker build -t ${{secrets.DOCKER_USERNAME}}/frontend:1.0 .
76
77 - name: Build backend image
78 working-directory: ./backend
79 run: docker build -t ${{secrets.DOCKER_USERNAME}}/backend:1.0 .
80
81 - name: Docker login
82 run: echo "${{secrets.DOCKER_PASSWORD}}" | docker login -u ${{secrets.DOCKER_USERNAME}} --password-stdin
83
84 - name: Push frontend
85 run: docker push ${{secrets.DOCKER_USERNAME}}/frontend:1.0
86
87 - name: Push backend
88 run: docker push ${{secrets.DOCKER_USERNAME}}/backend:1.0


added robust ci-cd - Sreevishn
github.com/Sreevishnu07/Microservices-Docker-CI-CD/actions/runs/24725788335/job/72327638380
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All jobs
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Run details
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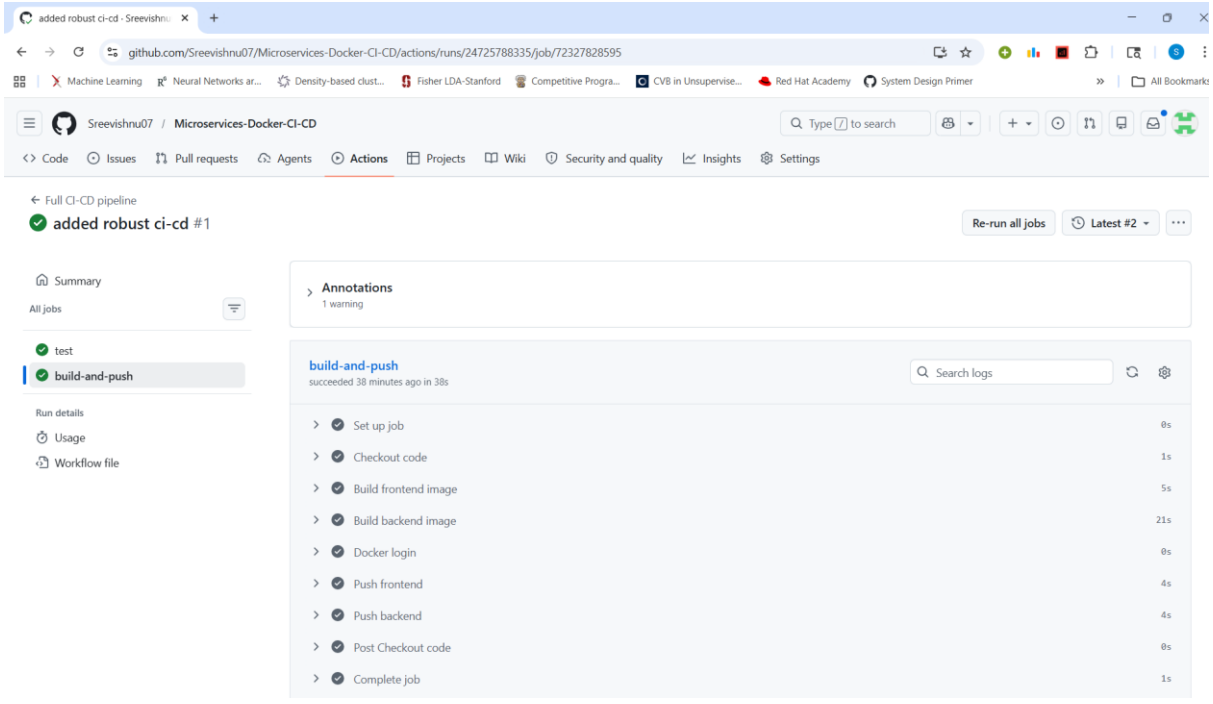
Workflow file
Annotations
1 warning
test
succeeded 39 minutes ago in 1m 1s
Search logs
Set up job 1s
Initialize containers 26s
Checkout code 1s
Setup Node 5s
Install backend dependencies 3s
Install MySql Client 10s
Wait for MySql 0s
Run Backend 10s
Test api 0s
Post Setup Node 0s
Post Checkout code 1s
Stop containers 1s
Complete job 0s


added robust ci-cd · Sreevishnu
github.com/Sreevishnu07/Microservices-Docker-CI-CD/actions/runs/24725788335/job/72327828595
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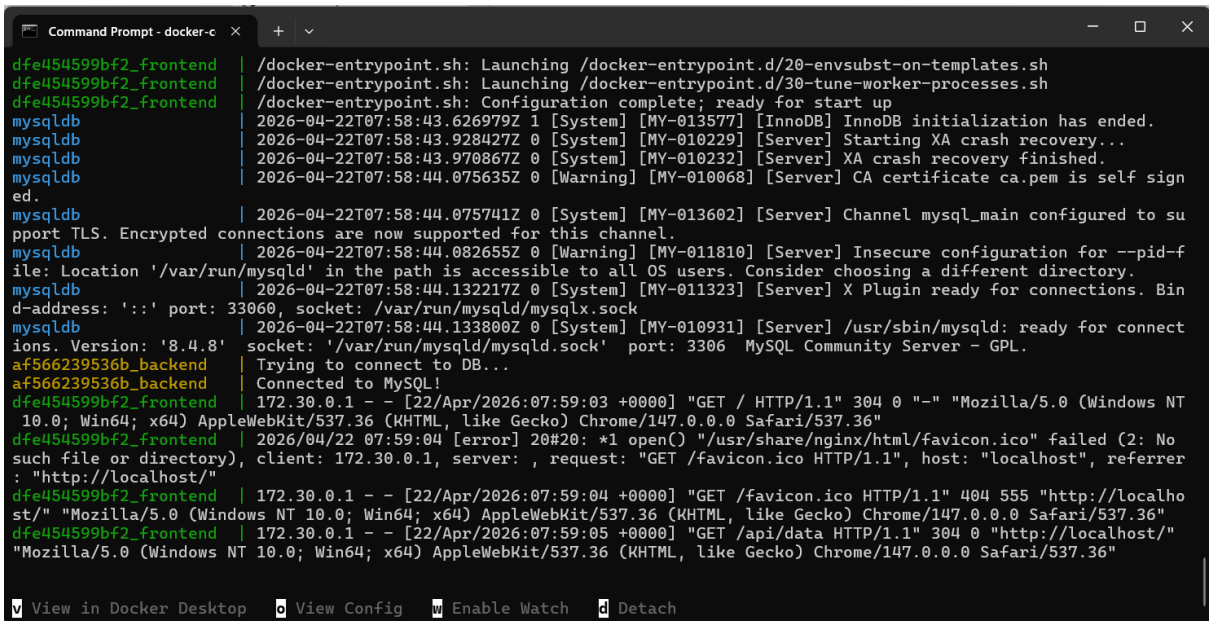
build-and-push
succeeded 38 minutes ago in 38s
Search logs
Set up job 0s
Checkout code 1s
Build frontend image 5s
Build backend image 21s
Docker login 0s
Push frontend 4s
Push backend 4s
Post Checkout code 0s
Complete job 1s


Command Prompt - docker-c
dfe454599bf2_frontend | /docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
dfe454599bf2_frontend | /docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
dfe454599bf2_frontend | /docker-entrypoint.sh: Configuration complete; ready for start up
mysqldb | 2026-04-22T07:58:43.626979Z 1 [System] [MY-013577] [InnoDB] InnoDB initialization has ended.
mysqldb | 2026-04-22T07:58:43.928427Z 0 [System] [MY-010229] [Server] Starting XA crash recovery...
mysqldb | 2026-04-22T07:58:43.970867Z 0 [System] [MY-010232] [Server] XA crash recovery finished.
mysqldb | 2026-04-22T07:58:44.075635Z 0 [Warning] [MY-010068] [Server] CA certificate ca.pem is self sign
ed.
mysqldb | 2026-04-22T07:58:44.075741Z 0 [System] [MY-013602] [Server] Channel mysql_main configured to su
pport TLS. Encrypted connections are now supported for this channel.
mysqldb | 2026-04-22T07:58:44.082655Z 0 [Warning] [MY-011810] [Server] Insecure configuration for --pid-f
ile: Location '/var/run/mysqld' in the path is accessible to all OS users. Consider choosing a different directory.
mysqldb | 2026-04-22T07:58:44.132217Z 0 [System] [MY-011323] [Server] X Plugin ready for connections. Bin
d-address: '::' port: 33060, socket: /var/run/mysqld/mysqlx.sock
mysqldb | 2026-04-22T07:58:44.133800Z 0 [System] [MY-010931] [Server] /usr/sbin/mysqld: ready for connect
ions. Version: '8.4.8' socket: '/var/run/mysqld/mysqld.sock' port: 3306 MySQL Community Server - GPL.
af566239536b_backend | Trying to connect to DB...
af566239536b_backend | Connected to MySQL!
dfe454599bf2_frontend | 172.30.0.1 - - [22/Apr/2026:07:59:03 +0000] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT
10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36"
dfe454599bf2_frontend | 2026/04/22 07:59:04 [error] 20#20: *1 open() "/usr/share/nginx/html/favicon.ico" failed (2: No
such file or directory), client: 172.30.0.1, server: , request: "GET /favicon.ico HTTP/1.1", host: "localhost", referrer
: "http://localhost/"
dfe454599bf2_frontend | 172.30.0.1 - - [22/Apr/2026:07:59:04 +0000] "GET /favicon.ico HTTP/1.1" 404 555 "http://localho
st/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36"
dfe454599bf2_frontend | 172.30.0.1 - - [22/Apr/2026:07:59:05 +0000] "GET /api/data HTTP/1.1" 304 0 "http://localhost/"
"Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36"
v View in Docker Desktop
o View Config
w Enable Watch
d Detach

Simple microservices app
localhost
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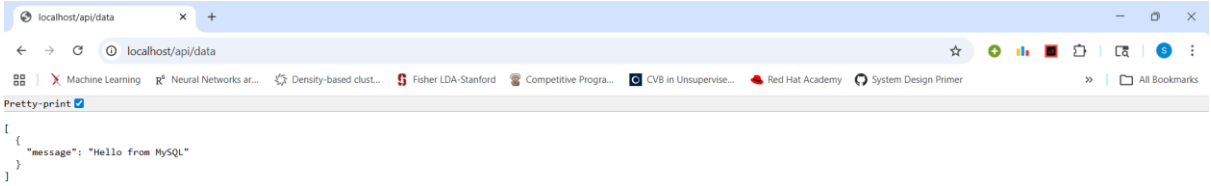
Click to get data
Get data




```
[
  {
    "message": "Hello from MySQL"
  }
]
```